UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
Lecturer: Eren C. Kızıldağ August 29, 2025
Scriber(s): Mohammad Alhyari

Proof of Lindeberg's CLT

Content.

1. Lindeberg Swapping.
2. Third Order Taylor Approximation
3. Lyapunov's Condition

Lindeberg's Central Limit Theorem (Lecture 2 Notes)

We continue the proof of Lindeberg's Central Limit Theorem.

Taylor Expansion

For a smooth function f, Taylor expansion around $y < x$ gives

$$f(x) = f(y) + f'(y)(x-y) + \frac{1}{2}f''(y)(x-y)^2 + \cdots + \frac{1}{N!}f^{(n)}(\xi)(x-y)^n,$$

for some $\xi \in [x, y]$.

Lindeberg's CLT

Theorem 1 (Lindeberg)**.** *Let X_{ij} satisfy the triangular array conditions (TAC). Then*

$$S_i = \sum_j X_{ij} \xrightarrow{d} N(0,1),$$

if the Lindeberg condition holds:

$$\forall\, \varepsilon > 0, \quad \sum_j \mathbb{E}\Big[X_{ij}^2\, \mathbf{1}_{\{|X_{ij}|>\varepsilon\}}\Big] \to 0.$$

Lindeberg Swapping Argument

We define partial sums with swapping:

$$\begin{aligned}
W_0 &= X_1 + X_2 + \cdots + X_{n-1} + X_n, \\
W_1 &= Z_1 + X_2 + \cdots + X_{n-1} + X_n, \\
W_2 &= Z_1 + Z_2 + X_3 + \cdots + X_{n-1} + X_n, \\
&\vdots \\
W_n &= Z_1 + Z_2 + \cdots + Z_{n-1} + Z_n,
\end{aligned}$$

where $Z_i \sim N(0, \sigma_i^2)$ are Gaussian random variables matched with X_i in the first two moments. Thus

$$W_n \sim N(0, \sum_j \sigma_j^2) \sim N(0, 1).$$

Convergence Criterion

From the previous lecture, recall: $W_0 \Rightarrow W_n$ if

$$\mathbb{E}[f(W_0)] \to \mathbb{E}[f(W_n)] \quad \text{for all } f \in C_b^{(3)}.$$

That is, for all bounded three times continuously differentiable functions f with

$$\sup_x |f(x)| \leq K, \qquad \sup_x |f^{(i)}(x)| < K_i, \quad i = 1, 2, 3.$$

Key Step

We want to bound the difference:

$$\Big|\mathbb{E}[f(W_{i-1})] - \mathbb{E}[f(W_i)]\Big|.$$

From last lecture we have:

$$|\mathbb{E}f(W_0) - \mathbb{E}f(W_n)| = \left|\sum_{i=0}^{n} \mathbb{E}f(W_i) - \mathbb{E}f(W_{i-1})\right| \overset{\Delta\text{-inequality}}{\leq} \sum_{i=0}^{n} |\mathbb{E}f(W_i) - \mathbb{E}f(W_{i-1})|$$

Now lets define the following random variables,

$$\begin{aligned}
W_{i-1} &= Z_1 + \cdots + Z_{i-1} + X_i + X_{i+1} + \cdots + X_n, \\
W_i &= Z_1 + \cdots + Z_{i-1} + Z_i + X_{i+1} + \cdots + X_n, \\
W_i^- &:= Z_1 + \cdots + Z_{i-1} + X_{i+1} + \cdots + X_n.
\end{aligned}$$

Now apply Taylor's expansion of f around W_i^-:

$$f(W_i) = f(Z_i + W_i^-) = f(W_i^-) + Z_i f'(W_i^-) + \frac{Z_i^2}{2} f''(W_i^-) + \frac{Z_i^3}{6} f^{(3)}(\gamma_1),$$

for some $\gamma_1 \in \{\min(W_i^-, W_i), \max(W_i^-, W_i)\}$.

$$f(W_{i-1}) = f(X_i + W_i^-) = f(W_i^-) + X_i f'(W_i^-) + \frac{X_i^2}{2} f''(W_i^-) + \frac{X_i^3}{6} f^{(3)}(\gamma_2),$$

for some $\gamma_2 \in \{\min(W_i^-, W_{i-1}), \min(W_i^-, W_{i-1})\}$.

Using the above Taylor expansions, we compute the difference:

$$\begin{aligned} f(W_i) - f(W_{i-1}) = (Z_i - X_i) f'(W_i^-) + \tfrac{1}{2}(Z_i^2 - X_i^2) f''(W_i^-) \\ + \tfrac{1}{6}(Z_i^3 f^{(3)}(\gamma_1) - X_i^3 f^{(3)}(\gamma_2)). \end{aligned}$$

Taking expectations,

$$\begin{aligned} \Big|\mathbb{E}[f(W_i)] - \mathbb{E}[f(W_{i-1})]\Big| = \Big|\mathbb{E}[(Z_i - X_i) f'(W_i^-)] + \frac{1}{2}\mathbb{E}[(Z_i^2 - X_i^2) f''(W_i^-)] \\ + \frac{1}{6}\mathbb{E}[Z_i^3 f^{(3)}(\gamma_1) - X_i^3 f^{(3)}(\gamma_2)]\Big|. \end{aligned}$$

By construction, X_i and Z_i match in the first two moments, so, using this and the fact that X_i, Z_i and W_i^- are independent, the first two terms vanish. Thus,

$$\Big|\mathbb{E}[f(W_i)] - \mathbb{E}[f(W_{i-1})]\Big| \leq \frac{1}{6}\Big(\mathbb{E}|X_i|^3 \sup_x |f^{(3)}(x)| + \mathbb{E}|Z_i|^3 \sup_x |f^{(3)}(x)|\Big).$$

Therefore,

$$\Big|\mathbb{E}[f(W_i)] - \mathbb{E}[f(W_{i-1})]\Big| \leq \frac{K}{6}\Big(\mathbb{E}|X_i|^3 + \mathbb{E}|Z_i|^3\Big),$$

where $K = \sup_x |f^{(3)}(x)|$

Theorem (Error Bound)

Summing over $i = 1, \ldots, n$, we obtain

$$\Big|\mathbb{E}[f(W_0)] - \mathbb{E}[f(W_n)]\Big| \leq \frac{K}{6} \sum_{i=1}^{n} \Big(\mathbb{E}|X_i|^3 + \mathbb{E}|Z_i|^3\Big).$$

Since $Z_i \sim N(0, \sigma_i^2)$, we use

$$\mathbb{E}|Z_i|^3 = \sigma_i^3 \mathbb{E}|N(0,1)|^3 = c\sigma_i^3.$$

Remark. Above, $c = \frac{2}{\sqrt{2\pi}} \int_0^\infty x^3 e^{-x^2/2}\, dx$, which is a finite constant.

Lyapunov Condition and Relation to Lindeberg

A useful trick:

$$\sigma_j^3 = (\mathbb{E}|X_j|^2)^{3/2}.$$

> **Fact (Monotonicity of Moments):** If $1 \leq p \leq q$, then
>
> $$(\mathbb{E}|X|^p)^{1/p} \leq (\mathbb{E}|X|^q)^{1/q}.$$
>
> This can be proven via, e.g., Jensen's inequality.

In particular,

$$(\mathbb{E}|X_j|^2)^{3/2} \leq \mathbb{E}|X_j|^3.$$

So, $\mathbb{E}|Z_i|^3 \leq c\sigma_i^3 \leq c\mathbb{E}|X_i|^3$ Thus,

$$\sigma_j^3 \leq C\mathbb{E}|X_j|^3.$$

Therefore,

$$\left|\mathbb{E}[f(W_0)] - \mathbb{E}[f(W_n)]\right| \leq \frac{K}{6}(c+1)\sum_{j=1}^{n}\mathbb{E}|X_j|^3.$$

So if $\sum_j \mathbb{E}|X_j|^3 \to 0$, then we are done.

Definition 1 (Lyapunov Condition)**.** *For $\delta > 0$, the Lyapunov condition holds with order δ if*

$$\sum_{j \leq n_i} \mathbb{E}|X_j|^{2+\delta} \to 0 \quad \textit{as } i \to \infty.$$

Thus, we are done if the Lyapunov condition holds with $\delta = 1$.

Remark. For the ordinary CLT, if $X_j = \frac{X-\mu}{\sigma\sqrt{n}}$ with X_j i.i.d., then

$$\sum_j \mathbb{E}|X_j|^3 = \sum_j \mathbb{E}\left|\frac{X-\mu}{\sigma\sqrt{n}}\right|^3 = \sum_j \mathbb{E}\left|\frac{X-\mu}{\sigma}\frac{1}{n\sqrt{n}}\right|^3 \sim O\left(\frac{1}{\sqrt{n}}\right).$$

So if $\mathbb{E}|X|^3 < \infty$, the classical CLT setting satisfies Lyapunov with $\delta = 1$, recovering a rate of order $O(1/\sqrt{n})$.

Example. Lyapunov $\Rightarrow$ Lindeberg.

Returning back, we would like to prove CLT under the Lindeberg condition.

Alternative Taylor Expansion

We can alternatively expand $f(W_i)$ as

$$f(W_{i-1}) = f(W_i^-) + X_i f'(W_i^-) + \frac{X_i^2}{2} f''(\gamma_3), \qquad \gamma_3 \in (W_i^-, W_{i-1}).$$

Notice that

$$\frac{X_i^2}{2} f''(W_i^-) + \frac{X_i^3}{6} f^{(3)}(\gamma_2) = \frac{X_i^2}{2} f''(\gamma_3),$$

where we matched our Taylor approximations to get this equation.

Thus, collecting terms,

$$\frac{X_i^3}{6} f^{(3)}(\gamma_2) = \frac{X_i^2}{2}\big(f''(\gamma_3) - f''(W_i^-)\big).$$

$$\frac{X_i^3}{6} f^{(3)}(\gamma_2) = \frac{X_i^2}{2}\big(f''(\gamma_3) - f''(W_i^-)\big)1_{\{|X_i|>\varepsilon\}} + \frac{X_i^3}{6} f^{(3)}(\delta_i)1_{\{|X_i|\le\varepsilon\}}$$

Therefore,

$$\begin{aligned}\Big|\mathbb{E}[f(W_i)] - \mathbb{E}[f(W_{i-1})]\Big| \le{}& \tfrac{K}{6}\,\mathbb{E}|Z_i|^3 \\ &+ \mathbb{E}\Big[\tfrac{X_i^2}{2}\,\big|f''(\gamma_3) - f''(W_i)\big|\,\mathbf{1}_{\{|X_i|>\varepsilon\}}\Big] \\ &+ \mathbb{E}\Big[\tfrac{1}{6}|X_i|^3 f^{(3)}(\gamma_2)\,\mathbf{1}_{\{|X_i|\le\varepsilon\}}\Big].\end{aligned}$$

Finishing the details, summing over j,

$$\begin{aligned}\Big|\mathbb{E}[f(W_0)] - \mathbb{E}[f(W_n)]\Big| \le{}& \underbrace{\tfrac{K}{6}\sum_j \mathbb{E}|Z_j|^3}_{(1)} \\ &+ \underbrace{\sum_j \mathbb{E}\Big[\tfrac{X_j^2}{2}\,(2K)\,\mathbf{1}_{\{|X_j|>\varepsilon\}}\Big]}_{(2)} \\ &+ \underbrace{\tfrac{K}{6}\sum_j \mathbb{E}\big[|X_j|^2\varepsilon\big]}_{(3)}.\end{aligned}$$

Now, notice the following:

$$\begin{aligned}(1) &= \tfrac{K}{6}\sum_j \mathbb{E}|Z_j|^3 = \tfrac{Kc}{6}\sum_j \sigma_j^3 \\ &\le \tfrac{Kc}{6}\Big(\max_{j\le n}\sigma_j\Big)\sum_j \sigma_j^2 \\ &= \tfrac{Kc}{6}\Big(\max_{j\le n}\sigma_j\Big)\cdot 1 \;\longrightarrow\; 0, \quad \text{by TAC.}\end{aligned}$$

Moreover, notice that (2) $\to 0$ by the Lindeberg condition,

$$\sum_j \mathbb{E}\big[X_j^2 1_{\{|X_j|>\varepsilon\}}\big] \to 0,$$

and finally $(3) \to \frac{K\varepsilon}{6}$ by TAC. Putting everything together:

$$\lim_{n\to\infty}\sum \mathbb{E}f(W_i) - \mathbb{E}f(W_{i-1}) \le \frac{K\varepsilon}{6}$$

Where ε is arbitrary; sending $\varepsilon \to 0$ we recover that $\mathbb{E}f(W_0) \to \mathbb{E}f(W_n)$.

We now give an example.

Example: $W_{n_i} \sim \mathrm{Bern}(p_n)$

Let

$$X_n = \frac{W_{n_i} - p_n}{\sqrt{p_n(1-p_n)}},$$

where X_n is the ith element of the jth row.

If $np_n(1-p_n) \to \infty$ (which holds if $\frac{1}{n} \ll p_n \ll 1 - \frac{1}{n}$), then Lindeberg's condition holds:

$$\sum_j \mathbb{E}\left[\frac{(W_{n_j}-p_n)^2}{np_n(1-p_n)}\mathbf{1}_{\{\,|W_{n_j}-p_n|>\varepsilon\sqrt{np_n(1-p_n)}\,\}}\right] \;\longrightarrow\; 0.$$

For any $\varepsilon > 0$, $\varepsilon\sqrt{np_n(1-p_n)} > 1$ for n large enough.

Berry–Esseen Theorem

Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. with

$$\mathbb{E}Y_i = 0, \qquad \mathbb{E}Y_i^2 = 1, \qquad \mathbb{E}|Y_i|^3 < \infty.$$

Then

$$\sup_{y\in\mathbb{R}}\left|\mathbb{P}\Big(\tfrac{1}{\sqrt{n}}\sum_{i=1}^n Y_i \le y\Big) - \mathbb{P}(N(0,1)\le y)\right| \;\le\; \frac{1.88}{\sqrt{n}}\,\mathbb{E}|Y_1|^3.$$

Observation

The Berry–Esseen theorem can be restated as

$$\sup_{f\in\mathcal{F}}\Big|\mathbb{E}[f(S_n/\sqrt{n})]-\mathbb{E}[f(N(0,1))]\Big| \;\le\; \frac{1.88}{\sqrt{n}}\,\mathbb{E}|Y_1|^3,$$

where $\mathcal{F}=\{1_{(-\infty,x]} : x\in\mathbb{R}\}$.

That is, for indicator functions,

$$\mathbb{P}(N(0,1)\le x)=\mathbb{E}[1_{\{N(0,1)\le x\}}].$$

Distances between Probability Measures

We define a general distance indexed by a class $\mathcal{D}$ of test functions:

$$d_{\mathcal{D}}(P,Q)=\sup_{f\in\mathcal{D}}\Big|\int f\,dP-\int f\,dQ\Big|=\sup_{f\in\mathcal{D}}\Big|\mathbb{E}_P[f(X)]-\mathbb{E}_Q[f(X)]\Big|.$$

Fact. If $\mathcal{D}$ is a separating class (namely $\mathbb{E}_P(f)=\mathbb{E}_Q(f)$ for all $f\in\mathcal{D}$ implies $P=Q$), then $d_{\mathcal{D}}$ is a proper metric.

Fact. $X_n \xrightarrow{d} X$ if and only if $d_{\mathcal{D}}(\mathcal{L}(X_n),\mathcal{L}(X))\to 0$ for $\mathcal{D}=\{1_{(-\infty,x]} : x\in\mathbb{R}\}$.

Total Variation Distance

$$d_{\mathrm{TV}}(P,Q)=\sup_{A\in\mathcal{B}}|P(A)-Q(A)|,$$

for probability measures P,Q defined on $(\mathbb{R},\mathcal{B})$.

However, total variation fails to capture convergence in distribution.

Example

Let X_j be i.i.d. Rademacher random variables with

$$\mathbb{P}(X_j=1)=\mathbb{P}(X_j=-1)=\tfrac{1}{2}.$$

Define

$$S_n=\sum_{j=1}^{n}X_j.$$

Then by the CLT,

$$\frac{S_n}{\sqrt{n}}\xrightarrow{d}N(0,1).$$

But in fact,

$$d_{\mathrm{TV}}\Big(\tfrac{S_n}{\sqrt{n}}, N(0,1)\Big) = 1,$$

since $\frac{S_n}{\sqrt{n}}$ takes values only in the discrete set

$$\mathcal{S}_n = \Big\{ \tfrac{k}{\sqrt{n}} : k = -n, -n+2, \ldots, n \Big\}.$$

For instance, take the set

$$A = \Big\{ \tfrac{1}{\sqrt{n}}\mathbb{Z} \Big\}^c,$$

then $\mathbb{P}\Big(\frac{S_n}{\sqrt{n}} \in A\Big) = 1$ but $\mathbb{P}(N(0,1) \in A) = 0$.

Exercise

Show that

$$X_n \xrightarrow{P} X \quad \not\Rightarrow \quad d_{\mathrm{TV}}(\mathcal{L}(X_n), \mathcal{L}(X)) \to 0,$$

but

$$d_{\mathrm{TV}}(\mathcal{L}(X_n), \mathcal{L}(X)) \to 0 \quad \Rightarrow \quad X_n \xrightarrow{P} X.$$